

INTERNSPARK TASK 1 REPORT

Name

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Task Title

Heart Disease Prediction Using Machine Learning

Objective

The objective of this project is to build a machine learning model that predicts whether a person is likely to have heart disease based on medical attributes.

Dataset Used

Heart Disease Dataset (heart.csv)

Tools and Technologies

- Google Colab
- Python
- Pandas
- NumPy
- Matplotlib
- Scikit-Learn

Methodology

Data Preprocessing

- Loaded the Heart Disease dataset.
- Checked dataset information and statistics.
- Handled missing values.
- Separated features and target variable.
- Standardized feature values using StandardScaler.

Model Building

Two machine learning models were implemented:

1. Logistic Regression
2. Random Forest Classifier

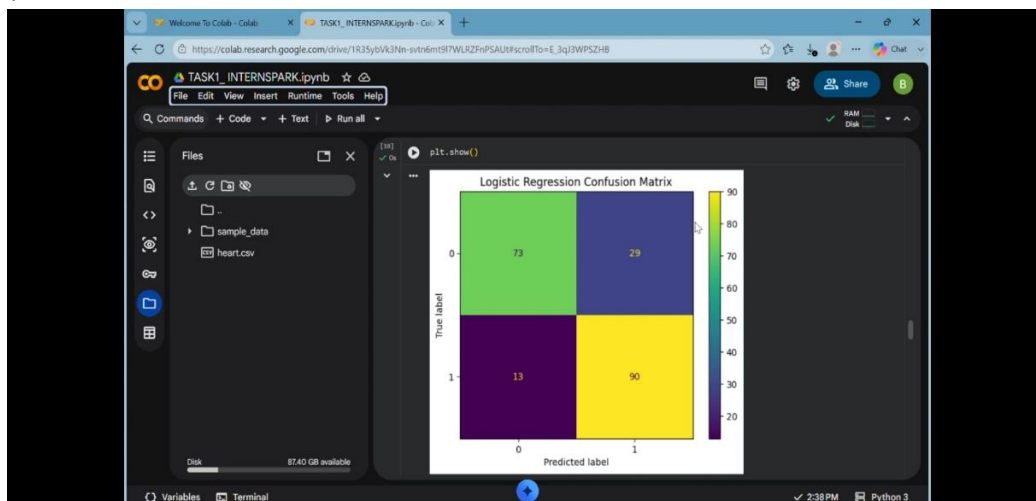
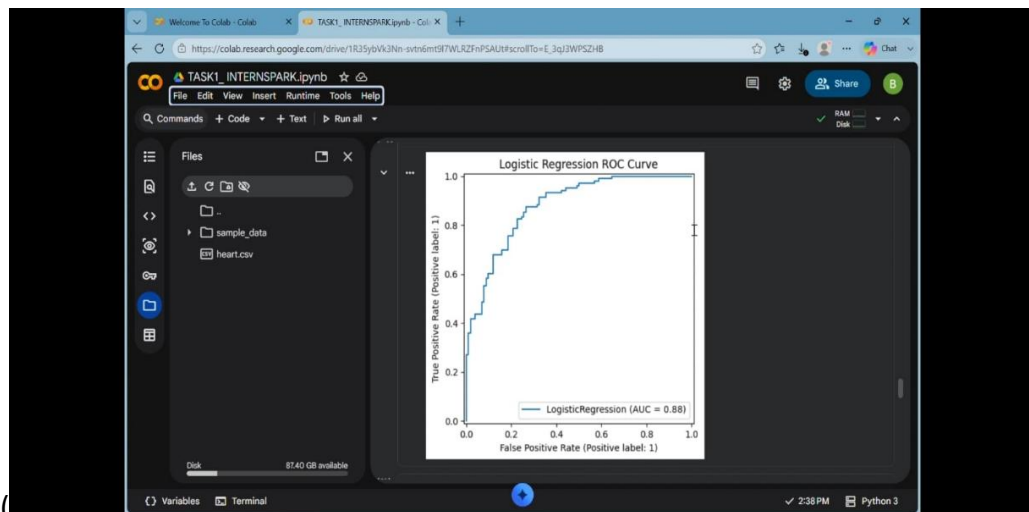
Model Evaluation

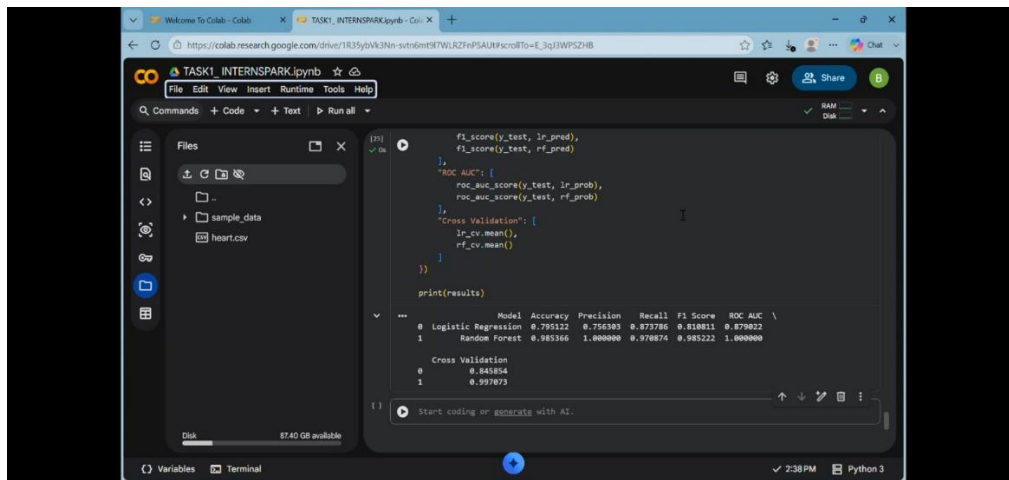
The models were evaluated using:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC Score
- Confusion Matrix
- ROC Curve

Results

Both models were trained and tested successfully on the dataset. Performance metrics were calculated and compared. ROC Curves and Confusion Matrices were generated for better evaluation of model performance.





GitHub Repository Link

<https://github.com/kundhana0712/InternSpark-Task1>

Conclusion

This project successfully implemented machine learning techniques for heart disease prediction. Logistic Regression and Random Forest models were compared using multiple evaluation metrics. The project demonstrated how machine learning can be applied to healthcare data for predictive analysis.